Getting started with spreadsheets.

1. Basic Computation

   (a) To add 3 + 4, enter `=3+4`

   (b) To subtract 100-76, enter `=100-76`

   (c) To multiply 4 times 18, enter `=4*18`

   (d) To divide 0.05 by 12, enter `=0.05/12`

   (e) To calculate 5^{25}, enter `=5^25`

	A	B
1	enter into a column with a leading =	answer
2	3+4	7
3	100-76	24
4	4*18	72
5	0.05/12	0.004166666667
6	5^25	2.98023E+17
7		

2. Compute an 18% tip on a $35.75 bill.

   (a) What is 18% as a decimal? 18/100 = 0.18

   (b) What should we enter into the spreadsheet? 35.75*0.18 ≈ $6.44

3. Make a tip calculator using **cell references**.

   (a) Open a new spreadsheet and label it `TIP`.

   (b) In cells A1, B1, C1, D1, type column headers:

`Bill Amount`	`Tip Percent`	`Tip Paid`	`Total`

   (c) In cells A2, B2, C2, D2, type:

$= 35.75$	$= 0.18$	$= A2 * B2$	$= A2 + C2$

   (d) What calculation did we do?

   C2 calculates the tip

   D2 calculates the total

   (e) Use the calculator to determine the Total if the Bill Amount is $76.80 and the Tip Percent is 20%.

   Change A2 to 76.80 & B2 to 0.2

   total = $92.16

4. Keeping the Tip Percent at 20%, we now demonstrate the **Fill Down** feature.

   (a) Enter 10 into cell `A2` and enter `=A2+10` into cell `A3`

   (b) Select `A3` and drag down until you get to 110.

   (c) Drag down cells `B2` through `D2`.

   (d) How much is the tip on a $110 meal? $22

   How much is the final bill? $132

   How large does the Bill Amount need to be for the tip to be more than $15? $80

   (the tip is $16)

	A	B	C	D
1	Bill Amount	Tip Percent	Tip Paid	Total
2	$10.00	0.2	$2.00	$12.00
3	$20.00	0.2	$4.00	$24.00
4	$30.00	0.2	$6.00	$36.00
5	$40.00	0.2	$8.00	$48.00
6	$50.00	0.2	$10.00	$60.00
7	$60.00	0.2	$12.00	$72.00
8	$70.00	0.2	$14.00	$84.00
9	$80.00	0.2	$16.00	$96.00
10	$90.00	0.2	$18.00	$108.00
11	$100.00	0.2	$20.00	$120.00
12	$110.00	0.2	$22.00	$132.00

5. Quick Summary of Finance Terms

(a) **principal**

Starting amount of money

also called present value

(b) **interest rate**

r = a percent that determines what percent of the principal should be added as interest

(c) **interest**

an amount of money added in each time period

(d) **accumulated amount** or **end amount** or **principal plus interest** or **future value**

$A = P + I$ = starting amount plus interest added

(e) **Formula for Simple Interest over Time**

$A = P + I = P + Prt = P(1+rt)$

P: principal; I: interest; Prt: how to calculate interest; P(1+rt): same, by algebra!

(f) Comment on time t.

interest can be added in using different time periods: annually, quarterly, monthly, etc.

6. Make a Simple Interest Calculator:

Suppose \$1000 is invested in a savings account that earns 3.2% interest on the \$1000 each year. ↑ principal ↑ interest rate

(a) Open a new spreadsheet/tab called `Interest`. Fill in the spreadsheet cells:

	A	B	C	D	E
1	Interest Rate	Principal	Year	Interest	Total
2	0.032	1000	0		=B2+D2
3	0.032	1000	=C2+1	=A3*B3	=D3+E2

(b) Use the pull down feature to determine the Total in the savings account in 10 years. Write down your steps.

Select A3 thru E3 & pull down until year = 10

(c) How long will it take for the savings account to double in value? Write down your steps.

We need total > 2000

Keep dragging! Eventually we see total = 2024 in year 32

(d) How can you calculate the Total in the savings account in 50 years **without** using the pull down feature?

If P is the principal, r is the interest rate, t is the number of years, and A is the accumulated amount of money in the savings account, write a formula for A. Then check your calculation using the pull down feature.

$A = P(1+rt) = 1000*(1+0.032*50) = 2600$

0.032	1000	48	32	2536
0.032	1000	49	32	2568
0.032	1000	50	32	2600
0.032	1000	51	32	2632

(e) A better simple interest calculator, using fixed cell references (the \$ fixes the row/column)

	A	B	C	D	E
1	Interest Rate	Principal	Year	Interest	Total
2	0.032	1000	0		
3			=C2+1	= A2*B2	=D3+E2

Now select and pull down `C3, D3, E3`. What happens if you change the interest rate to 4%? What if you change the principal to \$3000?

Everything updates w/o re-pulling!

7. How To Export the **entire workbook** as a **PDF**.

(a) Save your work!

(b) In upper left menu, click `File` to produce a dropdown menu.

(c) In dropdown menu, select `Download`

(d) In right-side menu, select `PDF`